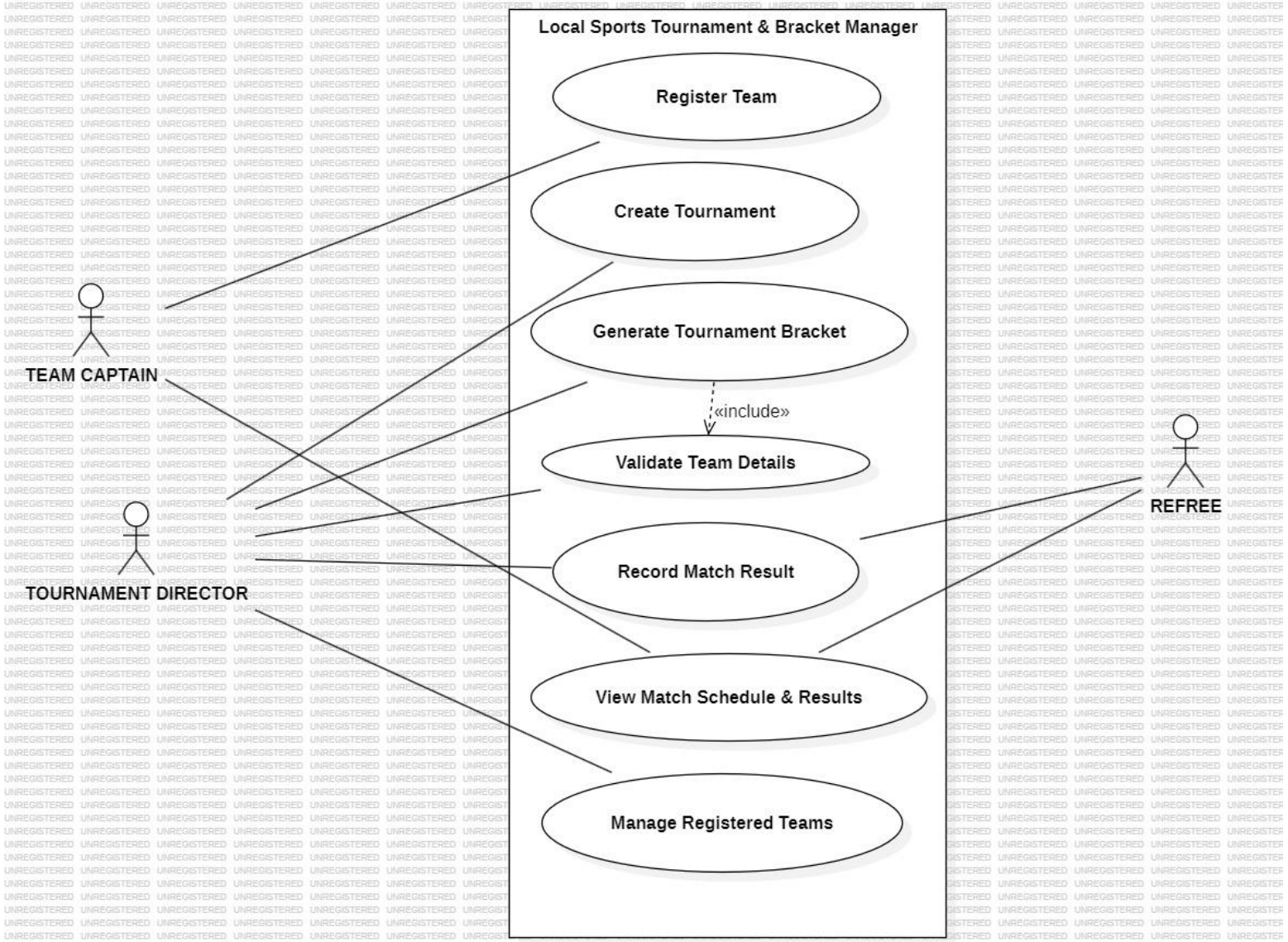
Local Sports Tournament & Bracket Manager

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall allow a Team Captain to register their team for an available tournament.	High	When valid team details are submitted, the system shall create the team registration and display a confirmation.	Core functionality for team participation.	Specify required team details and prevent duplicate registration.
FR-002	Functional	The system shall allow the Tournament Director to create a tournament by entering tournament details.	High	When valid tournament details are submitted, the system shall create and display the tournament.	Required to organize tournaments.	Specify mandatory tournament details such as name, sport, date, and format.
FR-003	Functional	The system shall allow the Tournament Director to generate a tournament bracket using the registered teams.	High	When at least 2 eligible teams are registered, selecting “Generate Bracket” shall create and display the bracket.	Essential for organizing matches.	Specify the minimum number of teams and supported bracket format.
FR-004	Functional	The system shall allow the Tournament Director or Referee to record the result of a completed match.	High	When a valid match result is submitted, the system shall save the result and update the tournament bracket.	Maintains accurate tournament progress.	Specify valid result format and who is authorized to record results.
FR-005	Functional	The system shall allow a Team Captain to view their team's match schedule and tournament results.	Medium	When the Team Captain selects their tournament, the system shall display the team's scheduled matches and recorded results.	Keeps teams informed about tournament progress.	Specify whether only the captain's own team informatio is displayed.
NFR-001	Nonfunc.	The system shall display the tournament bracket within 3 seconds after the user requests it.	High	In at least 95 out of 100 tests, the bracket shall appear within 3 seconds.	Ensures quick access during events.	Specify that the 3-second limit applies under normal system load.
NFR-002	Nonfunc.	The system shall prevent unauthorized users from modifying tournament information or match results.	High	A Team Captain attempting an unauthorized modification shall be denied access in all authorization tests.	Protects tournament data.	Specify access permissions for Team Captain, Tournament Director, and Referee.


Local Sports Tournament & Bracket Manager
Register Team
Create Tournament
Generate Tournament Bracket
«include»
Validate Team Details
Record Match Result
View Match Schedule & Results
Manage Registered Teams
TEAM CAPTAIN
TOURNAMENT DIRECTOR
REFREE

USER-CASE FLOW

Goal	Generate Tournament Bracket for Registered Teams
Actor	Tournament Director (primary); Team Captain, Referee (supporting)
Main flow	**1.** Tournament Director selects an existing tournament and chooses the **"Generate Tournament Bracket"** option.
	2. The system displays the list of registered teams for the selected tournament.
	3. The Tournament Director confirms the request to generate the bracket.
	4. The system validates the registered team details and checks that the teams are eligible for the tournament.
	5. The system checks that a sufficient number of eligible teams are available for bracket generation.
	6. The system generates the tournament bracket and assigns the registered teams to match slots.

7. The system displays the generated tournament bracket to the Tournament Director.

8. The Tournament Director reviews and confirms the generated bracket.

9. The system saves the tournament bracket and makes the match schedule available.

10. The tournament bracket generation is complete.

Alternate Flow

Insufficient Teams

At **Step 5**, if there are not enough eligible teams:

5a. The system displays an **"Insufficient Teams"** message to the Tournament Director.

5b. The system does not generate the tournament bracket.

5c. The Tournament Director is prompted to register additional teams.

5d. The use case ends.

Invalid Team Details

At **Step 4**, if one or more team details are invalid or incomplete:

4a. The system displays an error message identifying the invalid team details.

4b. The Tournament Director is prompted to correct the team information.

4c. The system validates the corrected information again.

4d. If the information is valid, the process continues from **Step 5**.